COMP1110 Computing and data science in everyday life

2026/27 Semester 1

Guidelines for Group Project

"Computing and Data Science in Everyday Life"

Introduction & Background

In a team of five, over two months, you will choose something from everyday life and study it using computing and/or data science.

This is a thinking and writing project, **not a software project**. You do not need to build a finished app. You do need an original question, a careful investigation, and a clear write-up. In outline, you will:

1. **Choose a topic.** Pick one specific, original question or situation from everyday life that computing or data science can help you study.
2. **Investigate it.** Read what others have already done, find a research gap (a part of the problem that has not been studied in depth), and suggest a computing or data science approach. That might be a data analysis, a model, an algorithm, or a system design.
3. **Present it.** Submit a Project Plan, a Final Report, a 10-minute group presentation, and more.

What is a "topic"? A topic is the question you are asking, not a tool and not a wide subject area. "Use ChatGPT", "build an app", and "apply machine learning" are tools. "Sports", "health", and "social media" are too broad. Most topics take one of these forms:

- a data question: what could we measure, and what would the analysis show?
- a design question: what tool or system would help here, and how would it work?
- a computational question: what algorithm, model, or simulation describes this situation?

Choose one of these as your main focus, at a level that is realistic for a first-year course. You do not need to invent a new algorithm, run a large study, or produce working software. Code and prototypes are optional and **not graded**. Diagrams, mock-ups, and screenshots are welcome if they help explain your idea.

We mainly assess the originality of your question and the quality of your reasoning. Look for a non-standard topic: not a common tutorial, and not a typical "beginner data science" exercise. Base it on a real setting, such as a hobby, a community, a local issue, the arts, a sport, or a niche activity. The project does not have to be conventionally "useful". Experimental, artistic, humorous, or simply curious ideas are fine, as long as there is real computing or data science thinking behind them.

Creativity Expectations

- The main creativity criterion is the originality of your topic, not the complexity of your solution.
- We reward topics that are:
    - Drawn from specific, concrete contexts (your community, hobbies, local issues, niche activities).
    - Unusual or surprising in how they connect everyday life to computing or data science.
    - Clearly the result of human curiosity and exploration, not just a standard assignment idea.
- The solution approach can be relatively standard (e.g., data collection, simple modeling, visualization). Creative solution designs will also be slightly rewarded (consider that as a small bonus), but are not the main focus of this project.

You may use AI to help brainstorm topics, but if many groups use similar prompts, you may end up with overlapping topics. If several groups choose essentially the same topic, this is a strong signal that it is not sufficiently creative. Topic overlap alone does not automatically mean low creativity; we will still judge the topic itself.

We acknowledge that creativity involves some subjectivity. Please do **not** email us to ask for an "estimated creativity score" for your topic. To prove creativity, you should undertake an extensive literature study and review of existing methods (and, where relevant, datasets and prior analyses). The point is to check that your problem or analysis has not already been studied in depth, and to identify a genuine **research gap**: a question, context, or angle that existing work leaves open. A topic that is well covered in published papers, textbooks, or standard tutorials is unlikely to meet the creativity criterion, even if it is new to you.

If your topic feels like something from a “Top 10 Data Science Projects for Beginners” post, it is likely not creative enough for this assignment.

Tips for finding a creative topic

- **Go narrow.** “Health” or “sport” is too broad. “How my climbing gym assigns belay partners” or “whether the campus bus is late in the rain” is closer to the right scale. Specific, local, and slightly obsessive is usually more original than important-sounding and generic.
- **Start from what you already care about.** Hobbies, sports, music, cooking, games, pets, family businesses, clubs, and volunteer work are good sources. You will notice questions that outsiders miss.
- **Follow the news, including science news.** A recent event, a local controversy, or an odd research story can suggest a question.
- **Ask people outside computing.** A musician, a shopkeeper, a coach, or a relative in another field can name problems you would not invent at a desk.
- **Then check the literature.** A topic that felt unique in conversation may already be a crowded research area. If the same question has been studied in depth, narrow the context, change the question, or move on.

Depth & Technical Substance

Your topic must be creative, but it also needs nontrivial technical depth in computing or data science. Projects that are purely ethical, social, historical, or philosophical, with no substantial computing or data science component, are **not suitable** for this assignment.

The project should have enough **substance and work** that five people can stay productively busy throughout the project period. A thin idea that one person could sketch in a few evenings will look under-scoped, even if it is original.

We are not counting hours or lines of code. Use this check instead: could each member point to a real share of the investigation, design, analysis, writing, or presentation that clearly needed time and thought? If the final report mainly restates a clever question with little developed follow-through, the project is too light for this assignment.

Your final report should show that you have engaged with the technical side at a level appropriate for a first-year computing and data science course. Detailed expectations by project type are at the end of this document.

In all cases, your final report should make it clear what the technical core is and demonstrate that you have engaged with it beyond a surface description. Projects that remain at the level of "idea + social discussion" without technical substance will not meet the requirements of this assignment.

Project Goals

- Identify a creative topic where computing or data science can be applied.
- Clearly describe the problem context and/or question you are exploring.
- Show how your topic connects to computing and data science concepts.
- Conduct a literature / background review to position your idea and identify a research gap.
- Reason about feasibility (data, methods, limitations).
- Work effectively as a team.
- Communicate your idea clearly in words.

Deliverables & Weighting

1. **Detailed Project Plan – 20%** (group assessment)
2. **Interim Progress Meeting – 3%** (group assessment)
    - a short meeting (~5 min) with the Student TA to report progress after Reading Week
3. **Interim Progress Log – 2%** (individual assessment)
    - a short written update (half a page) summarising your progress and contribution before the final submission
4. **Final Report – 45%** (group assessment)
5. **Group Presentation – 20%** (group + individual assessment)
    - 10 minutes per group, face-to-face
6. **Individual Self-Reflection – 10%** (individual assessment)
    - reflect on your personal learning, contributions, and views on creativity in computing and data science

Detailed guidelines and rubrics for each deliverable will be published in separate documents.

Group Size & Timeline

- Group size: 5 students

Due Dates (tentative – confirm on Moodle)

Project Plan	Oct 4
Interim Progress Meeting	Oct 20–24
Interim Progress Log	Nov 4
• Final Report • Individual Self-Reflection	Nov 15
Group Presentation	Nov 17/19/24/26

Free Riders

This is a group project, and every member is expected to contribute a fair share of the work. A **free rider** is someone who consistently does little or no work while relying on the rest of the group.

If you have a free-rider problem, or any other dispute about workload, please tell the teaching team. Do not wait until the marks are released if the problem is already clear. We would rather hear about it during the project, while there is still time to help.

At the end of the project, every group will complete a simple **peer review form**. This form is **not** used to adjust scores directly. It is only a way for us to spot possible free-rider cases, together with any reports you make during the project.

If a possible case comes up, we will investigate. Score changes are based on that investigation, not on the peer review numbers themselves. Where there is sufficient evidence, we may **adjust the scores** of free riders (and, if needed, of other group members). Honest,

specific reports help us do this fairly. Vague complaints with no examples are harder to act on.

Expected Depth by Project Type

Your project should show nontrivial technical depth in at least one of the following forms.

1. Development-Oriented Projects ("Build a Solution")

These projects propose a software tool, app, or system to address your chosen problem.

Expected depth:

- A clear problem specification that motivates the need for software.
- An app/system design: key screens or interfaces, main user flows, and core features.
- At least some of the following:
    - UI mock-ups or wireframes.
    - A simple architecture diagram (components, data flow, external services).
    - A feature list with priorities (must-have vs. nice-to-have).
- If you implement anything: brief description of technologies used and key design decisions.

You are not required to build a complete, production-ready system, but your design should be specific enough that a competent developer could understand what to build and why.

2. Data-Analysis / Data-Science Projects

These projects center on collecting, analyzing, or interpreting data related to your topic.

Expected depth:

- A clear description of:
    - What data you consider (type, source, how it would be collected or obtained).
    - Why this data is relevant to your problem.
- Some form of data analysis, such as:
    - Descriptive statistics (distributions, trends, correlations).
    - Simple visualizations (plots, maps, dashboards).
    - Basic modeling or pattern detection (e.g., clustering, simple regression, rule-based classification), even if done conceptually or with toy data.

The analysis does not need to be as sophisticated as in an advanced data science course, but it should go beyond "we would collect data" and show concrete analytical thinking.

3. Problem-Solving / Algorithmic Projects

These projects focus on solving a well-defined computational problem (e.g., a class of puzzles, optimization tasks, scheduling problems).

Expected depth:

- A precise problem definition:
    - Inputs, outputs, constraints, and success criteria.
- A proposed solution approach:
    - Algorithmic idea, heuristic, or strategy (even if not fully implemented).
    - Discussion of complexity or scalability at a high level (e.g., "this approach scales linearly with the number of items").
- If the problem is small enough to be tractable:
    - Include sample solutions or results (e.g., solved instances, performance on test cases).

Purely describing a puzzle or game without a serious computational framing is insufficient; there must be a clear computing or data science problem and solution concept.

4. Simulation / Modeling Projects

These projects propose a computational model or simulation of a real-world process (e.g., crowd movement, information spread, resource usage).

Expected depth:

- A clear description of:
    - The system or process being modeled.
    - The key variables, states, and rules in your model.
- A conceptual or simple simulation design:
    - How time steps, events, or interactions are handled.
    - What outputs or metrics the simulation produces.
- If possible:
    - Example simulation runs (even with small or synthetic data) and interpretation of results.

The focus is on showing that you understand how to translate a real-world phenomenon into a computational model, not on building a highly optimized simulator.

5. Hybrid Projects

Some projects will combine elements from the above types (e.g., a data-informed app design, or a simulation plus a proposed tool).

For hybrid projects:

- Clearly identify which parts are development, data analysis, algorithmic, or modeling.
- Ensure that at least one component has sufficient technical depth as described above.
- Avoid spreading yourself too thin; it is better to go moderately deep in one or two aspects than to be superficial in many.